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Bihar Engineering University, Patna

End Semester Examination - 2022

Time: 03 Hours

Code: 101507 ***Subject: Transportation Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer of the following (Any seven questions only): **[14]**

(a) The star and grid pattern of road network was adopted in
(i) Nagpur road plan
(ii) Lucknow road plan
(iii) Bombay road plan
(iv) Delhi road plan

(b) The desirable camber for straight roads with water bound macadam or gravel surface, is
(i) 1 in 33 to 1 in 25
(ii) 1 in 40 to 1 in 33
(iii) 1 in 150 to 1 in 140
(iv) 1 in 160 to 1 in 140

(c) IRC recommends the use of ___ curve as transition curve
(i) Spiral
(ii) Lemniscate
(iii) Cubic parabola
(iv) Square parabola

(d) The weaving length of a roadway is the distance
(i) between the channelizing islands
(ii) equal to half circumference
(iii) equal to total width of adjoining radial roads
(iv) equal to diameter of rotary

(e) Total parking demand is determined by which traffic study?
(i) Accident studies
(ii) Traffic volume study
(iii) Travel time studies
(iv) Parking studies

(f) Regulatory signs are generally circular in shape, identify a regulatory sign which is NOT circular in shape
(i) Restriction end sign
(ii) Speed limit sign
(iii) Stop sign
(iv) No parking sign

(g) The speed and delay studies on a defined section of highway are conducted by
(i) radar gun
(ii) traffic counters
(iii) moving car method
(iv) enoscope

(h) The penetration test for bitumen is conducted at a temperature of
(i) 60˚C
(ii) 37˚C
(iii) 25˚C
(iv) 50˚C

(i) The dowels bars are used in rigid pavements for
(i) Resisting bending stresses in all the portions
(ii) Transferring load from one portion to another
(iii) Resisting shear stresses in all the portions
(iv) Resisting Tensile stresses in all the portions

(j) Bitumen of grade 80/100 means
(i) Its penetration value is 10 cm
(ii) Its penetration value is 8 to 10 mm
(iii) Its penetration value is 8 to 10 cm
(iv) Its penetration value of 8 min

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Q.2 Solve both questions : **[14]**

(a) Differentiate between flexible and rigid pavement.

(b) Using the data given below, calculate the wheel load stresses at (a) interior (b) edge and (c) corner regions of a cement concrete pavement using Westergaard's stress equations. Also determine the probable location where the crack is likely to develop due to corner loading.
Wheel load, $P = 5100Kg$
Modulus of elasticity of cement concrete, $E = 3.1\times10^5 kg/cm^2$
Pavement thickness, h = 15cm
Poisson's ratio of concrete, $\mu = 0.15$
Modulus of subgrade reaction, $K = 6.0kg/cm^3$
Radius of contact area, a = 15cm

Q.3 Solve both questions : **[14]**

(a) On a two-way traffic road, the speeds of overtaking vehicles are 65 km/hr and 40 km/hr. If the average acceleration is $0.92m/s^2$. Determine the overtaking sight distance indicating the details of overtaking operations. Show the minimum length of overtaking zone and details of overtaking zone by a neat sketch.

(b) A valley curve of a state highway is formed by a descending gradient of 1 in 20 meeting an ascending gradient of 1 in 30. Design the length of a valley curve to fulfil both comfort condition and headlight sight distance required for a design speed of 80 km/hr. Assume allowance rate of change of centrifugal acceleration $C = 0.60m/s^3$. Suggest the best suitable shape of the valley curve. Consider reaction time t = 2.5sec and coefficient of friction = 0.35.

Q.4 Solve both questions : **[14]**

(a) Explain the factors affecting sight distances.

(b) The driver of a vehicle travelling 60 km/hr up a gradient requires 9 m less to stop after he applies brakes, as compared to a driver travelling at same speed, down the same gradient. Given, $f = 0.40$. What is the present gradient?

Q.5 Solve all questions : **[14]**

(a) Explain various road patterns in detail with neat sketches.

Q.6 Solve all questions : **[14]**

(a) Explain total reaction of driver and the factors on which it depends.

(b) Explain PIEV theory.

(c) Write a note on factors affecting friction offered by Pavement surface.

Q.7 Solve all questions : **[14]**

(a) Compare tar and bitumen.

(b) List the various tests carried out on bitumen.

(c) Explain the desirable properties of aggregate to be used in different types of pavement construction.

Q.8 Solve both questions : **[14]**

(a) Discuss the advantages and limitations of CBR method of design for flexible pavement.

(b) Discuss the critical combination of stresses due to wheel load and temperature effects.

Q.9 Write short notes on any four of the following: **[14]**

(i) Kerb Parking.

(ii) Design factors of highway lighting

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(iii) ESWL.

(iv) Traffic islands

(v) Off Tracking.

(vi) Super elevation

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